Proceedings of the International Conference on Artificial Intelligence and Smart Systems (ICAIS-2021)
IEEE Xplore Part Number: CFP21OAB-ART; ISBN: 978-1-7281-9537-7

2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS) | 978-1-7281-9537-7/20/$31.00 ©2021 IEEE | DOI: 10.1109/ICAIS50930.2021.9395753

Motor Imagery Classification Based on Hybrid Feature Extraction and Deep Neural Network

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***Abstract*—The brain-computer interface increases the capacity of critical diseases diagnosis in neurological science. The increased capacity of diagnosis uses motor imagery electroencephalogram signals. This paper prosed hybrid feature extraction methods based on wavelet packet decomposition and common spatial pattern. The hybrid techniques are very efficient instead of discrete wavelet transform and common spatial pattern. For the process of classification proposed deep neural network-based model. The proposed model enhances the classification ratio despite the machine learning algorithm. The method of classification ratio varies 8-12% instead of the machine learning algorithm. The multi-kernel approach of the deep neural network increases the sample size and boosts data mapping and performance of the classification algorithm. The proposed algorithm is tested on BCI competition-IV dataset (2a and 2b). The simulation process has used the MATLAB R2014a software; this software is a well-known algorithm analysis software. For the evaluation process of result, four parameters such as accuracy, sensitivity, specificity and error are measured. These parameters compare with existing methods of motor imagery EEG classification. The performance of parameters indicates the efficiency of proposed algorithm when compared with the existing methods.**

Keywords- BCI, MI, EEG, CSP, WPD, DNN, feature extraction, classification, MATLAB

I. INTRODUCTION

The brain computer interface (BCI) provides communication process of human brain control and outside device. The processing of brain computer interface in various domain such as medical rehabilitation, entertainment and many more visualization application of brain mapping of real-world environments[1, 2]. In area of medical rehabilitation, brain computer interface poses great potential in terms of recoding of human body activity signal record with the help of electrode [3]. The connection of electrode with slap has different position, most of authors use 22 electrode and 3 channels for the recoding of signals [4, 5]. The recoded signals have analog nature, after the recoding it convert into digital signal. The recoding and conversion of signals occupied with low signal to noise ratio (SNR) [6, 7]. The low signal to noise ratio deflects the classification process of motor imagery electroencephalogram (EEG). The structure of EEG data is very complex and the characteristic is non-stationary in domain of frequency and time. The motor imagery-based EEG classification process applied in various medical diagnosis for detection of critical disease, such as brain stroke, brain dead and epilepsy [8, 9, 10]. These critical diseases have very less recovery rate and finally the patient's loss their life. For the survival of life of patients is big issue in the area of biomedical diagnosis, so motor imagery classification plays great role in detection of critical disease [11, 12, 13]. For the detection of critical disease needs for the classification. The classification methods cannot directly apply for the process of classification in motor imagery classification process. The process of classification needs the process of feature extraction [14]. The process of feature extraction divides the different bands of signals in different range such as alpha, beta, gamma, delta and many more. These bands have certain frequency range for the mapping of motor imagery EEG signals [15]. The abnormal activity of human brains finds in these bands. The preprocessing and noise filtrations of motor imagery EEG signals is very crucial part. If the preprocessing of motor imagery is better the process of classification achieve the target ratio of classification. The conventional methods of feature extraction applied time and frequency-based methods for the measurement of energy factor and variation for the selection of bands. The conventional feature extraction methods decline the performance of classification algorithms. In survey various authors applied various methods for the extraction of feature components of EEG signals [16, 17]. The most of authors applied the common spatial pattern (CSP), discrete wavelet transform (DWT), AR, empirical wavelet transform (EWT) and wavelet packet decomposition (WPD) [18, 19, 20]. This paper applied hybrid feature extraction methods, the hybrid feature extraction methods used two function, one is common spatial pattern and other is wavelet packet decomposition. The hybrid feature extraction methods are very efficient instead of conventional and dynamic feature extraction methods [21, 22]. The classification accuracy depends on the handling of motor imagery EEG signals. The handling capacity is poor the classification ratio of motor imagery EEG classification suffers. The various authors reported support vector machine, CNN, machine learning, extreme learning algorithms, enhance the classification ratio. This paper proposed deep neural network-based classification algorithm [23]. The deep neural network classification algorithm is extension of multi-layer perceptron neural network. The complexity of DNN increase the classification ratio of motor imagery EEG signals [24, 25]. The rest of paper describe as in section II. Describe the related work. In section III describe proposed methodology. In section IV describe the experimental analysis and finally conclude in section V.

II. RELATED WORK

The classification accuracy is major issue in motor imagery EEG classification. For the enhancement of classification accuracy various authors proposed methods based on neural network, machine learning and many other approaches are used for the classification. Some significant methods describe here. Authors [1] describe the process of feature extraction methods based on common spatial pattern. The common spatial pattern enhances the process of features extraction of MI based EEG classification. Authors also applied the hybrid feature extraction methods along with different derivation of wavelet transform function. The extracted features improved the classification ratio of MI based EEG classification. In [2] authors proposed classification algorithms based on different classification such as KNN, SVM and decision tree. The methods of decision tree applied for the access of entropy-based feature selection methods. The process of feature selection reduces the artifices noise in signal of EEG data. In [3] authors proposed the diversion factor of feature represents of motor imagery-based EEG classification with SSDT transform. The process of transform divide in two categorized and process for the transformation of features. In [4] authors describe the process of feature extraction of MI based EEG data with WET transform function. The applied transform improves the classification of EEG signals. Authors also decreases the time complexity of algorithm; the proposed algorithm enhance the overall system of BCI. In [5] authors proposed the classification methods based on multiple channel selection methods; the process of methods is called MSFBCNN. The authors also describe the learning factor of proposed model of classifier. The mutual information of feature component is symmetrical of overall processing of features for the process of classification. In [7] authors proposed hybrid classification algorithm based on PCA and DBN. The PCA algorithm applied for the reduction of feature components of motor imagery-based EEG signals. The reduced feature matrics improve the classification accuracy and sensitivity of motor imagery data. In [8] authors proposed the CNN based classification algorithm. The proposed algorithm enhances the classification of EEG signals. The results of authors indicate the proposed algorithm applied for real time wearable device to control human emotion and others behaviors. [9] Authors have applied frequency decomposition methods for the selection of signals band in different categories. The derived frequency sampling applied on BCI competition IV dataset with all subjects. [10] Authors have applied 3-layer CNN classification methods for MI based EEG classification. The processing of data input applied soft-max margin methods and reduces the dimension of data. The reduced dimension of data overcome the limitation of training error. [11] Authors proposed cluster-based classification methods along with neural network models. The cluster-based algorithm derives the process of patterns of different bands of signals. The processing of feature matrix in combined formation of input vector. [12] The authors have proposed Bayesian based classification algorithm. The proposed algorithm also encapsulates the RVM methods of support vector. The RVM methods overcome the limitation of support vector machine. [15] Authors applied discrete wavelet transform methods for the extraction of feature components of EEG signals classification. The authors measure the movements of left and right hand of person. [17] Authors applied the classification algorithm and varies the ratio of cross-fold and increase the classification ratio of EEG signals. Also describe the methods of channel selection process. The channel selection process decreases the complexity of classifier. In [19, 20, 29] the authors applied the concept of micro-band selection for the MI based EEG classification. The applied feature extraction methods is wavelet transform with M=2. Here M represents the length of signals. [30] Authors have proposed the model of symbol training for the classification of MI based EEG classification. The major bottleneck problem of proposed algorithm is selection of bands during the process of classification and input vector represents.

III. PROPOSED METHODOLOGY

The proposed methodology describes in two section in one section describe the process of feature extraction based on hybrid feature extractor and in second section describe the process of classification with deep neural network. The process of feature extraction is important phase of motor imagery EEG classification [7, 9]. The process of feature extraction applied hybrid feature extraction method. The hybrid feature matrix is combination of common spatial patterns (CSP) and wavelet packet decomposition. The combined feature extraction methods increase the strength of feature extraction methods [11]. The process of CSP methods based on variance factor of matrix. And the wavelet packet decomposition is derived form of wavelet decomposition.

The transform of EEG data transforms in two-dimensional spatial sub space with projection matrix. The process of transform increases the variance of matrix. The CSP methods based on the continuous diagonalization of the covariance matrix of both classes.

The description of CSP algorithm, consider the movement of EEG signal as A and B, now the represents the signal as XA and XB with dimension M X T. where M is the number of channels and T is the number of samples per channels. Now the covariance of EEG as

$$RA = \frac{X_A \; XA^T}{trace(X_A \;\; X_A^T)}$$

$$RB = \frac{X_B \; XB^T}{trace(X_B \;\; X_B^T)} \; \ldots\ldots(1)$$

XT is the transpose of X and trace (A) estimate the total of the diagonal elements of A. the average normal covariance RA and RB is measure over all trail.

The composite spatial covariance is

$$R = RA + RB = \mathrm{U}\, o \sum \mathrm{U}_0^1 \; \ldots.(2)$$

Where U0 is the matrix of eigenvector and $\sum is$ diagonal matrix of eigenvector. The weighting transform matrix is

$$P = \sum^{-1/2} \mathrm{U}_0^1 \ldots \ldots..(3)$$

The wavelet packet decomposition is derived transform of wavelet transform. The derived transform overcome the limitation of wavelet decomposition. The wavelet decomposition divides the original signals into two subspaces V and W which is orthogonally complementary to each other. The wavelet decomposition form only the frequency axis of bands. The wavelet packet decomposition is advance version and form the tree of packet. The processing of hybrid feature extraction methods describe here. The process of hybrid feature extractor performed on Z channel of CSP. The formulation of channel describe in [26, 27, 28]. The process of feature extraction considers two feature components energy and standard deviation. The hybrid extractor selected sub-bands 256HZ and 250HZ. The selection of bands marks as dotted line.

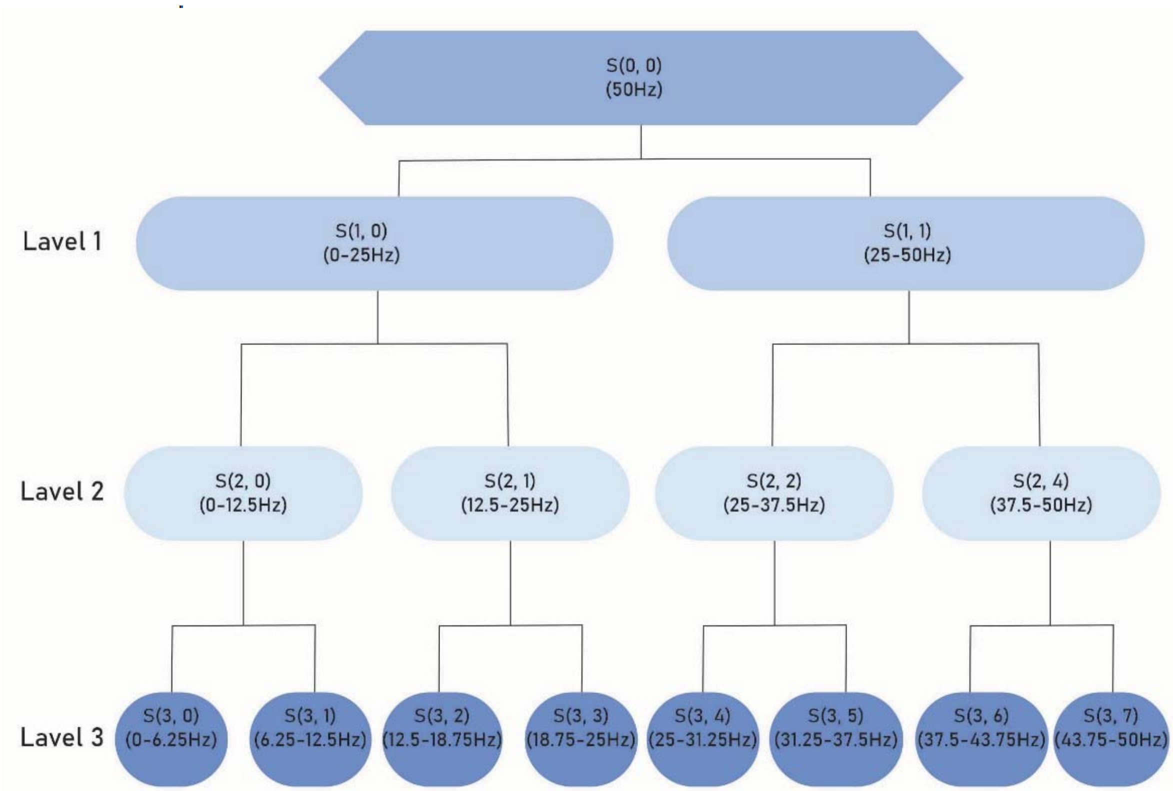

Figure 1. Shows the decomposition of signal with wavelet packet transform with level 3.

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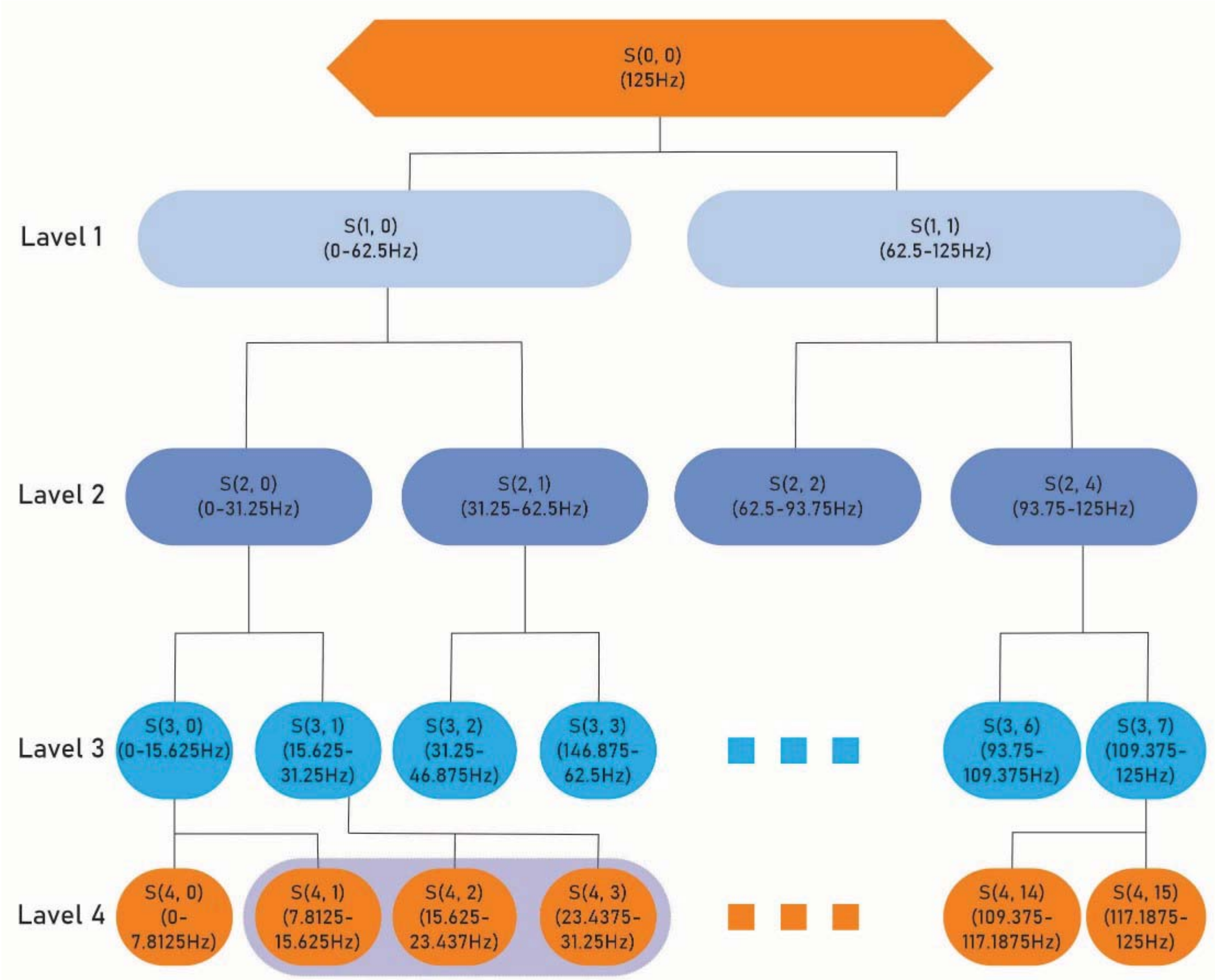

Figure 2. Shows the decomposition of signal with wavelet packet transform with level 4.

The estimation of the energy and standard derivation are the same as CSP-wavelet [12]. The feature vectors are

$$X_{hybrid} = \begin{cases} e_1^1, s_1^1, e_2^1, s_2^1 \\ e_1^2, s_1^2, e_2^2, s_2^2, \dots\dots, e_B^2, S_B^2 \ \ channel\ 2, \dots, e_B^1, s_B^1\, chnanel\ 1 \\ e_1^{2m}, s_1^{2m}, e_2^{2m}, s_2^{2m} \dots\dots\dots, e_B^{2m}, s_B^{2m} \qquad Channel\ 2m \end{cases}$$

Here B represent the selected sub-bands.

After the processing of feature extraction, consider the sample of feature components as $X = (x1, x2, x3, \dots\dots, xN)^T$ Where $X \in R^{N\,X\,P}$, the N is total number of feature components and P is dimensions of features sample.

The process of motor imagery EEG classification algorithm applied probability weight deep neural network (PWDNN). The proposed algorithm work in two section in forest section mapping the probabilistic function for the extracted feature components form hybrid feature extractor and in second section describe the processing of deep neural network. The deep neural network is derived form of multi-layer perceptron neural network. The proposed model applied 3 hidden layers. The process of probabilistic weight describe as the feature components value X= {x1, x2, x3 … xn}. The probability weight factor measures the covariance factor for the estimation of weight as

$$Cvi = \frac{n\sum x1x2 - \sum x1 \sum x2}{\sqrt[n]{(\sum x_1^2 - ((\sum x1)^2)(n\sum x_2^{2-(\sum x2)}}} \dots (4)$$

Now value of weight is calculating as

$$\delta i \longrightarrow \sum_{i=1}^{n} si \dots\dots (5)$$

Then maximize the feature components as

$$F^* = argmax\delta i\{x1, x2, \dots\dots, xn\} \dots (6)$$

The maximize weight consider as probabilistic function for the selection of input vector

The input layer of DNN is expressed by U as

$$u(t) = \sum_{i=1}^{N} Fi\delta_{uv} + \alpha_j \ \dots (7)$$

The equation (7) the input layer 'U' of Fi is feature components of motor imagery EEEG with wight 'δuv' and bias factor is $\alpha j.U(t)$.

The output of activation function is obtained by equation

$$\alpha = s1b1 + s2b2 + \cdots..+snbn \ldots (8)$$

The equation describes the relation of feature sub-bands and set factor of features b.

The output of hidden layer is described as

$$V_i(t) = \begin{cases} if\ ((s1,s2,\ldots sn) == b1,b2,\ldots..bn), retrun\ 1 \\ otherwise, \quad return\ o \end{cases} \ldots(9)$$

Here s1, s2,.,sn is feature components of motor imagery EEG and b1,b2,b3,……………..bn is predictor class.

The results obtained from this forwarded to weight adjustment factor as

$$w(t) = \delta_{uv} vi(t) \ldots\ldots\ldots\ldots (10)$$

After processing of classification data measure the error as

$$e(t) = \sum_{i=1}^{N} |\left(w(t)\right) - \overline{w(t)}\ | \ldots\ldots (11)$$

Here the $\overline{w(t)}$ is predicted sample and w(t) is label class of features of motor imagery EEG data.

Process of proposed Algorithm

Input: feature components F={x1, x2, x3 … .xn} with sample features s1, s2, s3 … sn

Output: classification ratio (AUC)

Begin

1. Define weight factor with probability weight function by equation (5)
2. For feature components sample $si\epsilon F$
3. While (e(t) is minimum) do
4. Call U(t) and process with hidden layer
5. Hidden layer wight w(t) with features components s
6. Output layer measure results
7. Estimate error
8. End
9. Exit

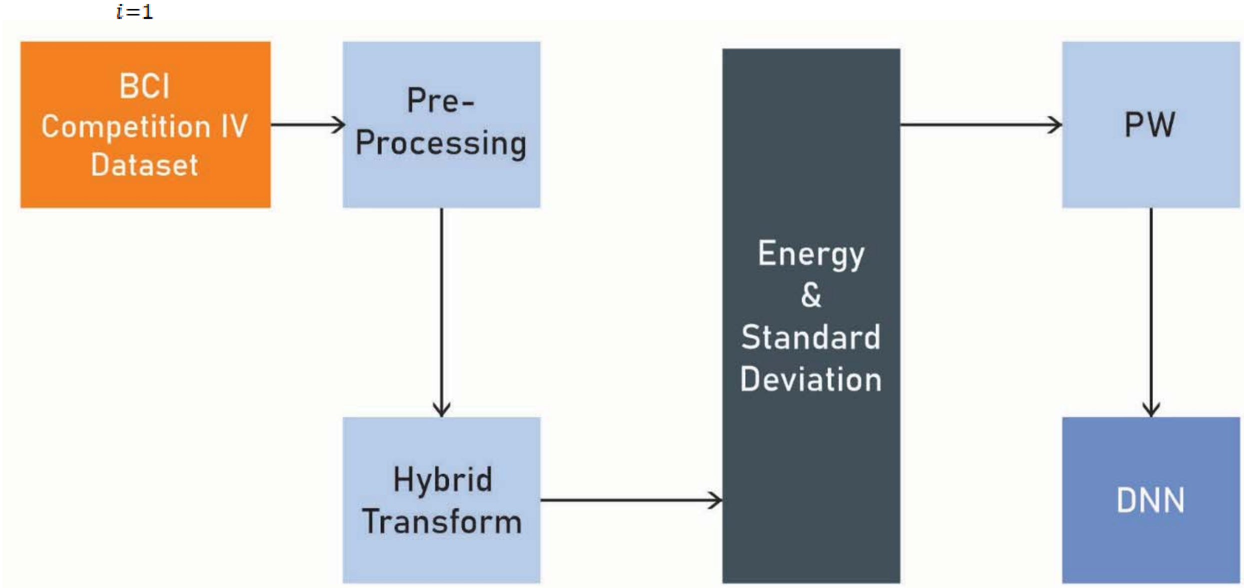

Figure 3. Process block diagram of proposed methodology based on hybrid feature extractor and deep neural network.

IV. EXPERIMENTAL ANALYSIS

To evaluate the performance of proposed algorithm for motor imagery EEG, applied BCI competition-VI dataset. For the process of simulation use MATLAB software with version of R2014a and the configuration of system is I9 processor, 16GB RAM with windows 10 operating system. The processing of sample data is 10 cross fold and measure these parameters accuracy, sensitivity, specificity and error [1, 2, 3]. The formula of parameters as

$$Accuracy = \frac{Total\ No.of\ Correctly\ Classified\ Instances}{Total\ No.of\ Instances} \times 100$$

$$Sensitivity = \frac{TP}{TP + FN} \times 100$$

$$Specificity = \frac{TN}{TN + FP} \times 100$$

TABLE I. COMPARISON ANALYSIS OF ACCURACY USING KELM, HKELM AND PROPOSED CLASSIFICATION WITH BC4 COMPETITION DATASET SUBJECT'S.

SUBJECT	KELM[18]	HKELM[18]	PROPOSED
S1	56.295	57.515	81.598
S2	74.234	78.469	79.278
S3	89.541	78.485	91.465
S4	69.564	74.247	76.756
S5	85.425	89.514	93.423
S6	66.477	69.525	75.145
S7	84.131	85.436	86.721
S8	77.524	78.932	81.554
S9	68.497	70.712	74.283

TABLE II. COMPARISON ANALYSIS OF SENSITIVITY USING KELM, HKELM AND PROPOSED CLASSIFICATION WITH BC4 COMPETITION DATASET SUBJECT'S.

SUBJECT	KELM[18]	HKELM[18]	PROPOSED
S1	82.979	89.178	90.498
S2	75.664	84.498	85.474
S3	84.331	86.651	89.116
S4	78.413	81.136	84.625
S5	80.746	83.994	86.634
S6	80.279	75.149	81.164
S7	84.582	85.266	92.928
S8	78.893	76.331	80.622
S9	76.471	75.574	78.416

TABLE III. COMPARISON ANALYSIS OF SPECIFICITY USING KELM, HKELM AND PROPOSED CLASSIFICATION WITH BC4 COMPETITION DATASET SUBJECT'S.

SUBJECT	KELM[18]	HKELM[18]	PROPOSED
S1	84.696	77.436	87.985
S2	78.345	80.485	84.632
S3	82.935	85.174	88.330
S4	75.674	84.616	88.234
S5	84.318	86.659	90.544
S6	78.416	81.148	83.868
S7	88.913	90.973	92.984
S8	78.259	80.681	83.616
S9	73.594	78.491	79.372

TABLE IV. COMPARISON ANALYSIS OF ERROR RATE (%) USING KELM, HKELM AND PROPOSED CLASSIFICATION WITH BC4 COMPETITION DATASET SUBJECT'S.

SUBJECT	KELM[18]	HKELM[18]	PROPOSED
S1	0.84	0.87	0.65
S2	0.78	0.85	0.68
S3	0.82	0.89	0.78
S4	0.84	0.70	0.65
S5	0.84	0.80	0.78
S6	0.80	0.72	0.61
S7	0.88	0.84	0.73
S8	0.68	0.64	0.59
S9	0.73	0.78	0.66

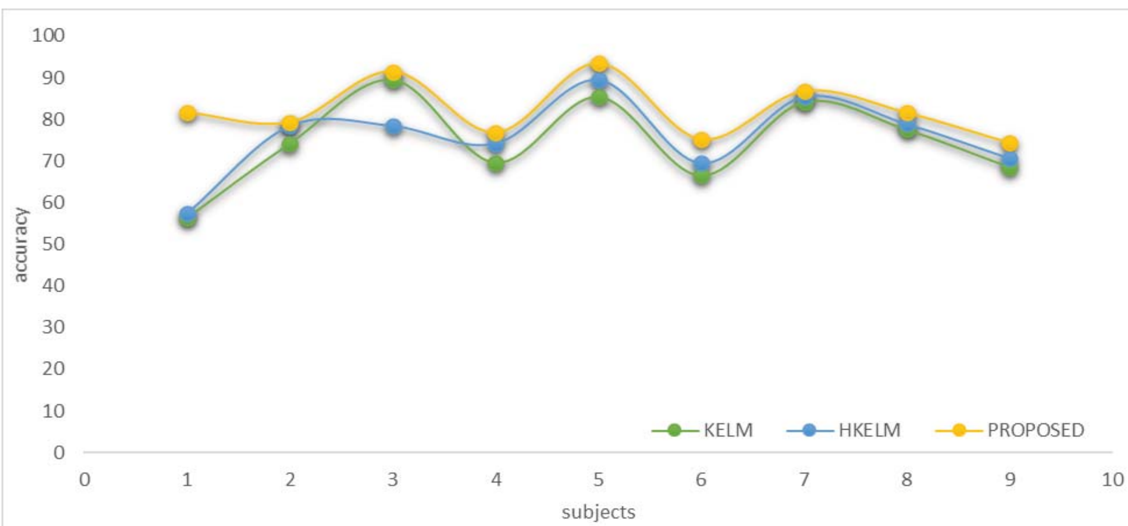

Figure 4. Comparison analysis of accuracy using KELM [18], HKELM[18] and PROPOSED classification with BC4 competition dataset subject's. The performance of proposed technique indicate with yellow color, HKELM technique indicate with blue color and KELM technique indicated with green color. Here we observed with each subject point of datasets the performance of proposed technique shown the best accuracy (in percentage) in comparison of other both two techniques. The proposed technique is accuracy variation 70 to 92% in the above graph.

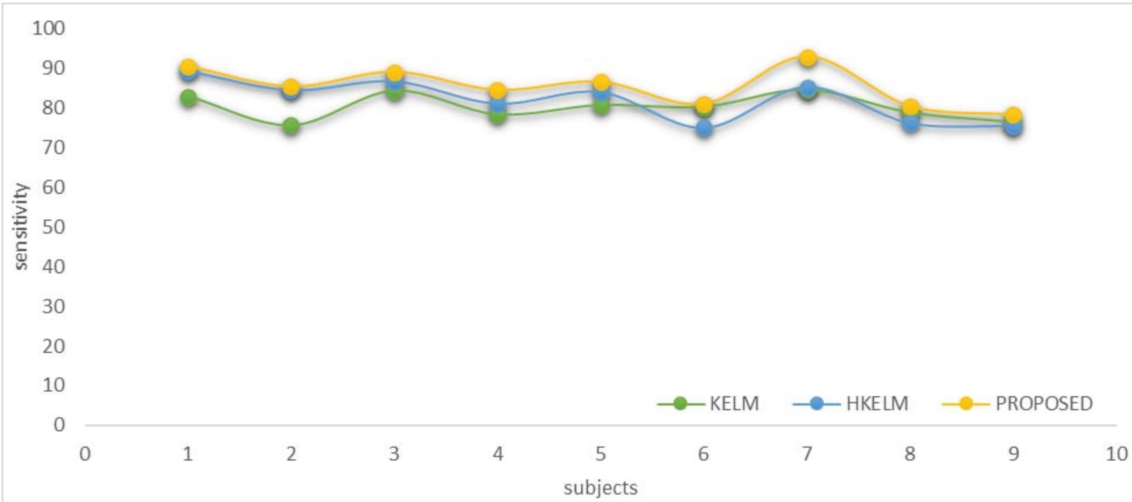

Figure 5. Comparison analysis of sensitivity using KELM [18], HKELM [18] and PROPOSED classification with BC4 competition dataset subjects. The performance of proposed technique indicate with yellow color, HKELM technique indicate with blue color and KELM technique indicated with green color. Here we observed with each subject point of datasets the performance of proposed technique shown the best sensitivity (in percentage) in comparison of other both two techniques. The proposed technique is sensitivity variation 75 to 95% in the above graph.

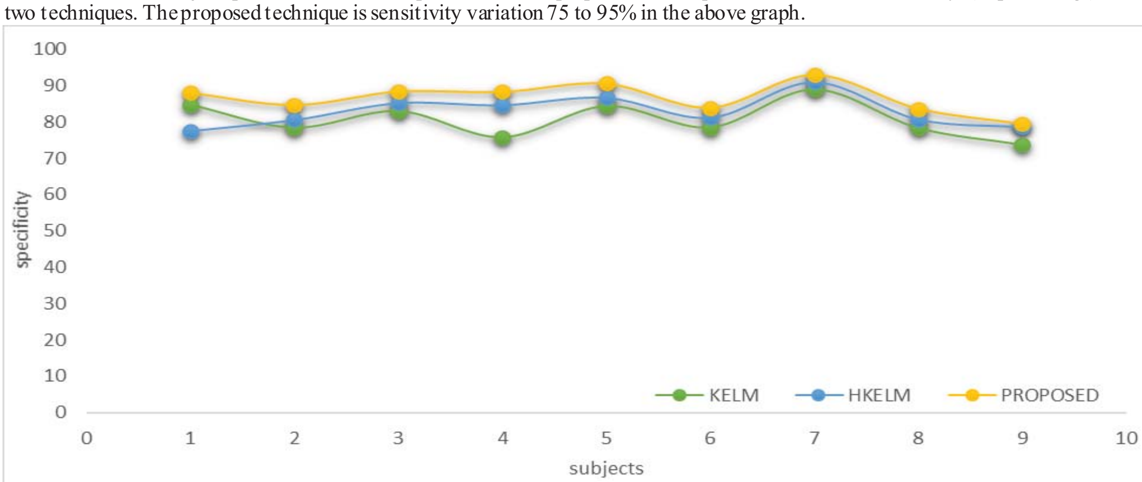

Figure 6. Comparison analysis of specificity using KELM [18], HKELM [18] and PROPOSED classification with BC4 competition dataset subjects. The performance of proposed technique indicate with yellow color, HKELM technique indicate with blue color and KELM technique indicated with green color. Here we observed with each subject point of datasets the performance of proposed technique shown the best specificity (in percentage) in comparison of other both two techniques. The proposed technique is specificity variation 78 to 96% in the above graph.

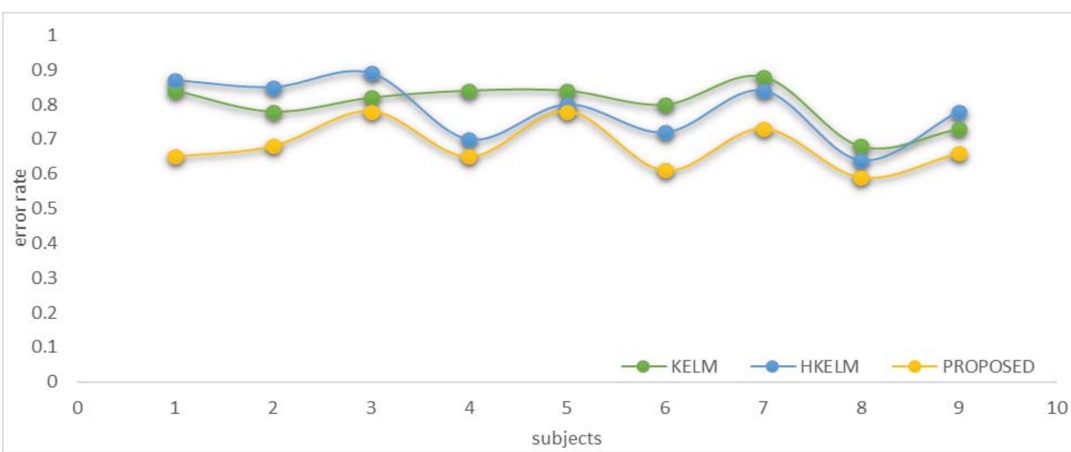

Figure 7. Comparison analysis of error rate (%) using KELM [18], HKELM [18] and proposed classification with BC4 competition dataset subjects. The performance of proposed technique indicate with yellow color, HKELM technique indicate with blue color and KELM technique indicated with green color. Here we observed with each subject point of datasets the performance of proposed technique shown the best specificity (in percentage) in comparison of other both two techniques. The proposed technique is specificity variation 78 to 96% in the above graph.

V. CONCLUSION & FUTURE SCOPE

This paper proposed probability-based weight function as a feature selector for deep neural network (DNN) classification of motor imagery EEG classification. The proposed methods mapped the feature space data to the sub-space matrix of features extractor. For the extraction of features, components applied hybrid features transform, and the hybrid transform combines common spatial patterns and wavelet packet decomposition. The hybrid features extractor efficient extract the feature components in terms of sub-bands. The increased value of accuracy, sensitivity and specificity indicated the proposed algorithm is better than existing algorithms. Also, measure the training error rate of the sample and the reduced error rate indicates the better training process of motor imagery EEG data classification. The probability weight function derives the random weight value function, the function value reparative, so uncounted feature component is repeated. Now in future optimized these feature components.

ACKNOWLEDGEMENT

I would like to thank Mr. Amit Kolhe, Managing Trustee of the Sanjivani College of Engineering, Kopargaon, India and the Dean, Faculty of Engineering, MPU, Bhopal also the HOD of Computer Department, MPU, Bhopal for providing the infrastructure required to carry out the proposed research work.

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